

For Animal Use Only

# Metamox 50% w/w

## Soluble Powder

**CONTENT :** Each gm contains:  
Amoxicillin trihydrate.....575 mg  
(Equivalent to Amoxicillin 500 mg)

**PRODUCT DESCRIPTION:** A white or almost white color powder.

### PHARMACODYNAMICS:

Amoxicillin is a time-dependent bactericidal antibiotic belonging to the semisynthetic penicillin group which acts by inhibiting the synthesis of bacterial cell walls during bacterial replication. It has a broad spectrum of activity against Gram positive and Gram negative bacteria, and owes its activity to the inhibition of the development of the peptidoglycan network structure in the bacterial cell wall. There are three main mechanisms of resistance to beta-lactams: beta-lactamase production, production of penicillin binding proteins (PBP), and decreased penetration of the outer membrane. One of the most important is the inactivation of penicillin by beta-lactamase enzymes produced by certain bacteria. These enzymes are capable of cleaving the beta-lactam ring of penicillins, making them inactive. The beta-lactamase could be encoded in chromosomal or plasmidic genes. Cross-resistance is observed between amoxicillin and other penicillins, particularly with aminopenicillins. Observed resistance rates are variable.

### PHARMACOKINETICS:

Amoxicillin is well absorbed following oral administration and it is stable in the presence of gastric acids. Excretion of amoxicillin is mainly in the unchanged form via the kidneys to give high concentration in renal tissue and urine. Amoxicillin is well distributed in body fluids. Studies in birds have indicated that amoxicillin is distributed and eliminated more rapidly than in mammals. Biotransformation appeared a more important route of elimination in birds than in mammals.

### INDICATION:

Treatment of infections in chickens, turkeys and ducks caused by bacteria susceptible to amoxicillin.

Pigs: For the treatment of pasteurellosis caused by *Pasteurella multocida* susceptible to amoxicillin.

### DOSAGE & ADMINISTRATION:

For use in drinking water.

Prepare the solution with fresh potable water immediately before use. Any medicated water which is not consumed within 24 hours should be discarded and the medicated drinking water replenished. In order to ensure consumption of the medicated water, animals should not have access to other water supplies whilst being treated.

The following formula may be used to calculate the required concentration of product (in milligrams of product per litre of drinking water):

$$\frac{\text{X mg product per kg bodyweight per day}}{\text{mean daily water consumption (l) per animal}} \times \frac{\text{mean bodyweight (kg) of animals to be treated}}{1} = \text{X mg product per litre drinking water}$$

The maximum solubility of the product is 65 g/L.

To ensure a correct dosage, bodyweight should be determined as accurately as possible to avoid underdosing. The uptake of medicated water depends on the clinical condition of the animal. In order to obtain the correct dosage the concentration of amoxicillin has to be adjusted taking into account water intake. After the end of the medication period the water supply system should be cleaned appropriately to avoid intake of sub-therapeutic amounts of the active substance.

#### Chickens:

The recommended dosage is 15 mg amoxicillin trihydrate (equivalent to 13.1 mg amoxicillin) per kg bodyweight per day (corresponding to 27 mg product/kg bodyweight/day)  
The total period of treatment should be for 3 days or in severe cases for 5 days.

#### Ducks:

Recommended dosage is 20 mg amoxicillin trihydrate (equivalent to 17.4 mg amoxicillin)/kg bodyweight per day (corresponding to 35 mg product/kg bodyweight/day) for 3 consecutive days.

#### Turkeys:

Recommended dosage is 15-20 mg amoxicillin trihydrate (equivalent to 13.1 - 17.4 mg amoxicillin)/kg bodyweight per day (corresponding to 27-35 mg product/kg bodyweight/day) for 3 days or in severe cases for 5 days.

#### Pigs:

Administer in the drinking water to give 20 mg amoxicillin trihydrate (equivalent to 17.4 mg amoxicillin)/kg bodyweight (corresponding to 35 mg product/kg bodyweight) daily.

The dose should be divided and administered at approximately 12 hourly intervals for up to 5 days.

The calculated dose should be measured out with calibrated scales.

### CONTRAINDICATIONS:

Do not use in rabbits, hamsters, gerbils and guinea pigs, or to birds producing eggs intended for human consumption.

Not effective against beta-lactamase producing organisms.

Do not use in known cases of hypersensitivity to penicillin or other substances of the beta-lactam group or to any of the excipients.

### WARNINGS / PRECAUTIONS:

Pigs: The uptake of medication by animals can be altered as a consequence of illness. In case of insufficient uptake of water, animals should be treated parenterally.

Special precautions for use

Special precautions for use in animals

Not effective against beta-lactamase producing organisms. Official, national and regional antimicrobial policies should be taken into account when the product is used.

Use of the product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria.

Special precautions to be taken by the person administering the veterinary medicinal product to animals

Penicillins and cephalosporins may cause hypersensitivity (allergy) following injection, inhalation, ingestion or skin contact.

Hypersensitivity to penicillins may lead to cross reactions to cephalosporins and vice versa. Allergic reactions to these substances may occasionally be serious.

1) Do not handle this product if you know you are sensitised or if you have been advised not to work with such preparations.

2) Handle this product with great care to avoid exposure, taking all recommended precautions.

3) If you develop symptoms following exposure such as a skin rash, you should seek medical advice and show the doctor this warning. Swelling of the face, lips or eyes or difficulty with breathing are more serious symptoms and require urgent medical attention.

• Avoid inhalation of dust.

• Wear gloves during preparation and administration of medicated water.

• Wash any exposed skin after handling the product or medicated water.

• Wash hands after use.

### DRUGS INTERACTIONS:

Amoxicillin exerts its bactericidal action by inhibition of bacterial cell wall synthesis during multiplication. It is therefore in principle not compatible with bacteriostatic antibiotics (e.g. tetracyclines, macrolides and sulphonamides) which inhibit multiplication.

Synergism occurs with  $\beta$ -lactam antibiotics and aminoglycosides.

### PREGNANCY AND LACTATION:

The safety of the veterinary medicinal product has not been established during pregnancy and lactation. Laboratory studies in rats and rabbits have not produced any evidence of teratogenic, foetotoxic or maternotoxic effects.

Use only according to the benefit/risk assessment by the responsible veterinarian.

### ADVERSE EFFECTS:

Penicillins and cephalosporins may cause hypersensitivity following administration. Allergic reactions to these substances may occasionally be serious.

### OVERDOSE AND TREATMENT:

In case of overdosing the treatment should be symptomatic. No specific antidote is available.

**STORAGE CONDITIONS:** Store at room temperature below 30°C. Protect from heat and sunlight exposure.

### WITHDRAWAL PERIOD(S):

Chickens (meat & offal): 1 day

Ducks (meat & offal): 9 days

Turkeys (meat & offal): 5 days

Pigs (meat & offal): 2 days

Not for use in birds producing or intended to produce eggs for human consumption.

**DOSAGE FORMS AND PACKAGING AVAILABLE:** 100gm water soluble packed into Aluminium foil.

**DATE OF REVISION OF PACKAGE INSERT:** 31 July 2018



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